

# Foundations of Machine Learning (IT582)

1st In-Semester Exam

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Total Marks: 50

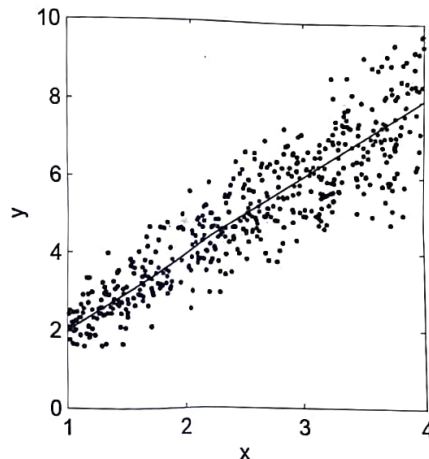
Duration: 2 Hrs.

All The Best!!

- Let us consider a regression model where the noise variance is a function of the input (variance increase as a function of input). Specifically

$$y = wx + \epsilon$$

where the noise  $\epsilon$  is normally distributed with mean 0 and standard deviation  $\sigma x$ . The value of  $\sigma$  is assumed to be known and the input  $x$  is restricted to the interval  $[1, 4]$ . We can write the model more compactly as  $y \sim \mathcal{N}(wx, \sigma^2 x^2)$ . If we let  $x$  vary within  $[1, 4]$  and sample outputs  $y$  from this model with some  $w$ , the regression plot might look like follows.



How is the ratio  $y/x$  distributed for a fixed  $x$ ? Suppose, we now have  $n$  training examples  $\{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\}$ , where each  $x_i$  is chosen at random from  $[1, 4]$  and the corresponding  $y_i$  is subsequently sampled from  $y_i \sim \mathcal{N}(w^* x_i, \sigma^2 x_i^2)$  with some true underlying parameter value  $w^*$ ; the value of the  $\sigma^2$  is the same as in our model.

- What is the maximum-likelihood estimate of  $w$  as a function of training data?
- What is the variance of this estimator due to the noise in the target outputs as a function of  $n$  and  $\sigma^2$  for fixed inputs  $x_1, x_2, \dots, x_n$ ?

[Marks 10]

- Consider the following dataset for predicting house prices using linear regression. The size of the house  $x_1$  is expressed in 1000 ft<sup>2</sup>, and the price  $y$  is expressed in \$1000.

| Size $x_1$ (1000 ft <sup>2</sup> ) | No. of bedrooms $x_2$ | Price $y$ (\$1000) |
|------------------------------------|-----------------------|--------------------|
| 1                                  | 1                     | 200                |
| 2                                  | 1                     | 250                |
| 1                                  | 2                     | 230                |
| 2                                  | 2                     | 300                |

Assume the linear regression model and find out the following.

- Construct the design matrix  $X$ , including the bias term.
- Using the normal equation, calculate the regression parameters  $\hat{\theta}$ .
- Use the obtained regression model to predict the price of a house having an area of 1500 and 4 bedrooms.

[Marks 10]

3. Consider  $n$  successive independent tosses of a coin. Let  $\theta$  denote the probability of obtaining a Head in a single toss. For the  $i$ -th toss, define

$$y_i = \begin{cases} 1, & \text{if the outcome is Head,} \\ 0, & \text{if the outcome is Tail.} \end{cases}$$

Assume that each observation follows a Bernoulli distribution. Further, assume that the prior distribution of  $\theta$  is a Beta distribution:

$$P(\theta) = \frac{1}{B(\alpha, \beta)} \theta^{\alpha-1} (1-\theta)^{\beta-1}, \quad \alpha, \beta > 1.$$

where

$$B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha + \beta)}$$

is the Beta function, and  $\Gamma(\cdot)$  is the Gamma function. For positive integers  $n$ ,  $\Gamma(n) = (n-1)!$ .

- Derive the likelihood function  $P(\mathbf{y} | \theta)$  for the  $n$  independent coin tosses.
- Using the MAP estimation criterion

$$\hat{\theta}_{MAP} = \arg \max_{\theta} P(\theta | \mathbf{y}),$$

derive an expression for  $\hat{\theta}_{MAP}$  by maximizing the log-posterior distribution. Express your answer in terms of  $y_1, y_2, \dots, y_n, \alpha$ , and  $\beta$ .

- Show that for  $\alpha = \beta = 1$ , the Beta prior becomes a uniform distribution over  $[0, 1]$ . Further, show that the MAP estimate reduces to the maximum likelihood (ML) estimate  $\hat{\theta}_{ML}$ .

[Marks 10]

4. While travelling in a bus, Ramesh saw a beautiful scene  $\mathbf{X}$ . As a photo enthusiast, he wanted to capture the scene using his mobile camera. However, due to the movement of the bus, the captured image  $\mathbf{Y}$  was blurred. To recover the original image, Ramesh modeled the image formation process for a small image patch as

$$\mathbf{y} = \mathbf{B}\mathbf{x} + \mathbf{n},$$

where  $\mathbf{B}$  is the blurring matrix and  $\mathbf{n}$  represents the noise introduced during image acquisition. Ramesh further assumed that an original image patch  $\mathbf{x}$  can be represented as a linear combination of patches from a collection of images. He therefore constructed a dictionary matrix  $\mathbf{A}$  whose columns contain image patches, such that

$$\mathbf{x} = \mathbf{A}\boldsymbol{\theta}.$$

- (a) Ramesh first estimates  $\theta$  by minimizing the least-squares data fidelity term. Derive the least-squares estimate  $\hat{\theta}$  and hence obtain an expression for the reconstructed image patch  $\hat{x}$  in terms of  $A$ ,  $B$ , and  $y$ .
- (b) Ramesh observes that the reconstructed image contains undesirable artefacts, particularly in smooth regions such as the sky. To suppress these artefacts, he introduces a smoothness regularization term based on the squared  $L_2$  norm of the Laplacian of the image patch. Derive the expression for the regularized reconstructed image patch  $\hat{x}$  in terms of  $A$ ,  $B$ ,  $L$ ,  $y$ , and  $\lambda$ .

[Marks 10]

5. Consider the following quadratic objective function:

$$J(\theta) = \frac{1}{2} \theta^T \begin{bmatrix} 5 & 2 \\ 2 & 2 \end{bmatrix} \theta - \begin{bmatrix} 7 \\ 4 \end{bmatrix}^T \theta,$$

where

$$\theta = \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix}.$$

- (a) Find the gradient  $\nabla_{\theta} J$  and the Hessian matrix  $H$ .
- (b) Find the stationary point  $\theta^*$  by solving

$$\nabla_{\theta} J = 0.$$

- (c) Starting from

$$\theta^{(0)} = \begin{bmatrix} 0 \\ 0 \end{bmatrix},$$

perform two iterations of gradient descent using  $\eta = 0.2$ .

- (d) Calculate

$$J(\theta^{(0)}), \quad J(\theta^{(1)}), \quad J(\theta^{(2)}).$$

Comment on whether the objective function is decreasing during the iterations.

- (e) Compare  $\theta^{(2)}$  with the stationary point  $\theta^*$  and explain why they are different.

[Marks 10]